Calibration Client Network Protocol

This document defines the communications protocol for the ColourSpace Calibration Client operation.

The protocol enables the definition of patch size, shape, position, and more the external patch generators, as used with the ColourSpace Java App, and the LightSpace Connect Android and iOS App, as well as other 3rd party systems.

Network

The communication between server and client occurs over the TCP/IP stacks and it's using sockets. Once the network interface is active on the ColourSpace application one socket is opened on port (default 20002) listening for connecting clients.

Create a connection

Connecting is easy; just open a stream on the address and port of the computer running the ColourSpace application when the network interface is started.

Getting the data

Every message is anticipated by an integer which contains the size of the message, if the size is a negative number it means that the communication is over and the client can close the connection.

The next step is to build the buffer to contain the message, read and parse the XML data from the stream to display the shapes and the colour on the screen.

Java code sample

```java
while (true)
{
  if (socket != null)
  {
    // create a buffer using the size of the message
    mexLen = input.readInt();
    if (mexLen <= 0)
    {
      // if negative, it's the disconnect message
      System.out.println("Lost connection with the server");
      closeConnection();
    }
    else
    {
      char[] dataBytes = new char[mexLen];
      // get the real data
      reader.read(dataBytes);
      // parse the XML
      xmlDoc = parser.build(new String(dataBytes), "");
      System.out.println("Message received”);

      // draw the shapes on the UI

.....

    }
  sleep(1000);

} // while
```

XML Message

Calibration

The root XML node.

Shapes

Child of the calibration node, it contains all the shapes that we want to display.

Rectangle

One of the possible shapes (and the only one supported). The “shapes” node can contain multiple rectangles. It supports these mandatory child nodes.

Color

The colour that fills the area of the rectangle. It has 3 attributes: “red”, “green” and “blue” and each of them is an integer value in the interval 0 to 255.

Colex

The extended colour that fills the area of the rectangle. It has 4 attributes: “bits”, “red”, “green” and “blue”. “bits” is a value from the integer range 8 to 16 inclusive, which can be used to determine the maximum value of the following colour integers. “red”, “green” and “blue” each have an integer value in the interval 0 to (2^bits) - 1.

Geometry

The geometry node describes the position and the size of the rectangle. There are 4 attributes: the values in “x” and “y” set the position of the top-left corner while “cx” and “cy” contains the size on the X and Y axis. Each attribute is a relative double between 0.0 and 1.0 and must be converted to the actual size of the screen.

Multiple Messages

The protocol will send one or more <rectangle> data blocks. These should be rendered in the order they are received. This mechanism is specifically used when calibration patch is not full screen and there is a non-black (0,0,0) background. The background <rectangle> is sent first, followed by the calibration <rectangle>. It is assumed that all <rectangles> are rendered onto a full screen default of (0,0,0).

ColourSpace Broadcast Identification

In order to simplify the process of connection, the current running instance of ColourSpace network server will broadcast its presence with a simple message on port 20123.
The message is in plain text and is of the following format:

LS:"network server ip address":"network server port number"
Eg LS:192.168.0.2:12321

Sample XML

Sample XML message to describe a full screen red rectangle, with default 8 bit colour data and extended 10 bit colour data

```
<?xml version="1.0" encoding="UTF-8" ?>
<calibration>
    <shapes>
        <rectangle>
            <color red="255" green="0" blue="0" />
            <colex bits=”10”red="1023" green="0" blue="0" />
            <geometry x="0.0" y="0.0" cx="1.0" cy="1.0"/>
        </rectangle>
    </shapes>
</calibration>
```

Sample XML message to describe a small white rectangle on a red background, with default 8 bit colour data and extended 10 bit colour data

```
<?xml version="1.0" encoding="UTF-8" ?>
<calibration>
    <shapes>
        <rectangle>
            <color red="255" green="0" blue="0" />
            <colex bits=”10”red="1023" green="0" blue="0" />
            <geometry x="0.0" y="0.0" cx="1.0" cy="1.0"/>
        </rectangle>
        <rectangle>
            <color red="255" green="255" blue="255" />
            <colex bits=”10”red="1023" green="1023" blue="1023" />
            <geometry x="0.25" y="0.25" cx="0.5" cy="0.5"/>
        </rectangle>
    </shapes>
</calibration>
```